

Computational Financial Modelling

IBEX-35 Quantitative Analysis

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1 Introduction

This report summarizes the Assignment-2 assigned to B23406 and being performed on the IBEX-35 data as we used the same in the last Assignment(Assignment-1).

The analysis includes:

- Data verification and cleaning
- Exploratory data analysis
- Normality tests
- Correlation Analysis
- Random portfolio Generation
- Markowitz Portfolio Optimization

All computations and visualizations were implemented using Python.

2 Data Verification and Cleaning

The dataset comprises historical daily stock prices from June 2010 to December 2025 for 19 selected IBEX 35 stocks, totaling approximately 75,000 dates.

2.1 Missing Values

A check for missing values was performed across all columns including Date, Open, High, Low, Close, and Volume. No missing values were detected in the dataset.

NaN Check Summary					
Date	0	Open	0	Ticker	0
Close	0	Volume	0	Stock	0
High	0	Company	0		
Low	0				dtype: int64

2.2 Duplicate Records

Duplicate entries were examined using the pandas `duplicated()` function. The dataset contains zero duplicate records, indicating that each observation is unique.

Duplicates Summary	
Duplicates:	0

2.3 Date Ordering

The data was sorted by stock ticker and date to ensure chronological ordering.

```
1 data = data.sort_values(["Stock", "Date"])
```

Sorted Records

Date	Close	High	Low	...	Volume	Stock
2025-12-18	3.480000	3.503000	3.453000	...	27606425	Telefónica
2025-12-19	3.420000	3.467000	3.395000	...	56409660	Telefónica
2025-12-22	3.419000	3.423000	3.368000	...	19546829	Telefónica
2025-12-23	3.436000	3.468000	3.402000	...	12480301	Telefónica
2010-06-01	11.920112	30.179079	29.081839	...	1052145	ACS
2010-06-02	11.863087	30.154367	29.170803	...	618915	ACS
2010-06-03	11.900447	30.485518	29.778734	...	787966	ACS
2010-06-04	11.597628	30.099998	28.716091	...	1263824	ACS
2010-06-07	11.499309	29.576092	28.552988	...	910833	ACS

[75315 rows x 9 columns]

This guarantees correct temporal structure for time-series analysis.

2.4 Price Adjustment Verification

To verify whether stock prices were adjusted for splits or dividends, extreme daily returns were inspected. No abnormal jumps in returns were observed, indicating that the dataset likely uses adjusted prices.

2.5 Trading Day Continuity

The difference between consecutive trading dates was computed. The median gap between observations was one day, with occasional gaps of up to five days corresponding to weekends and public holidays. This confirms that the dataset follows the normal trading calendar.

Continuity Statistics

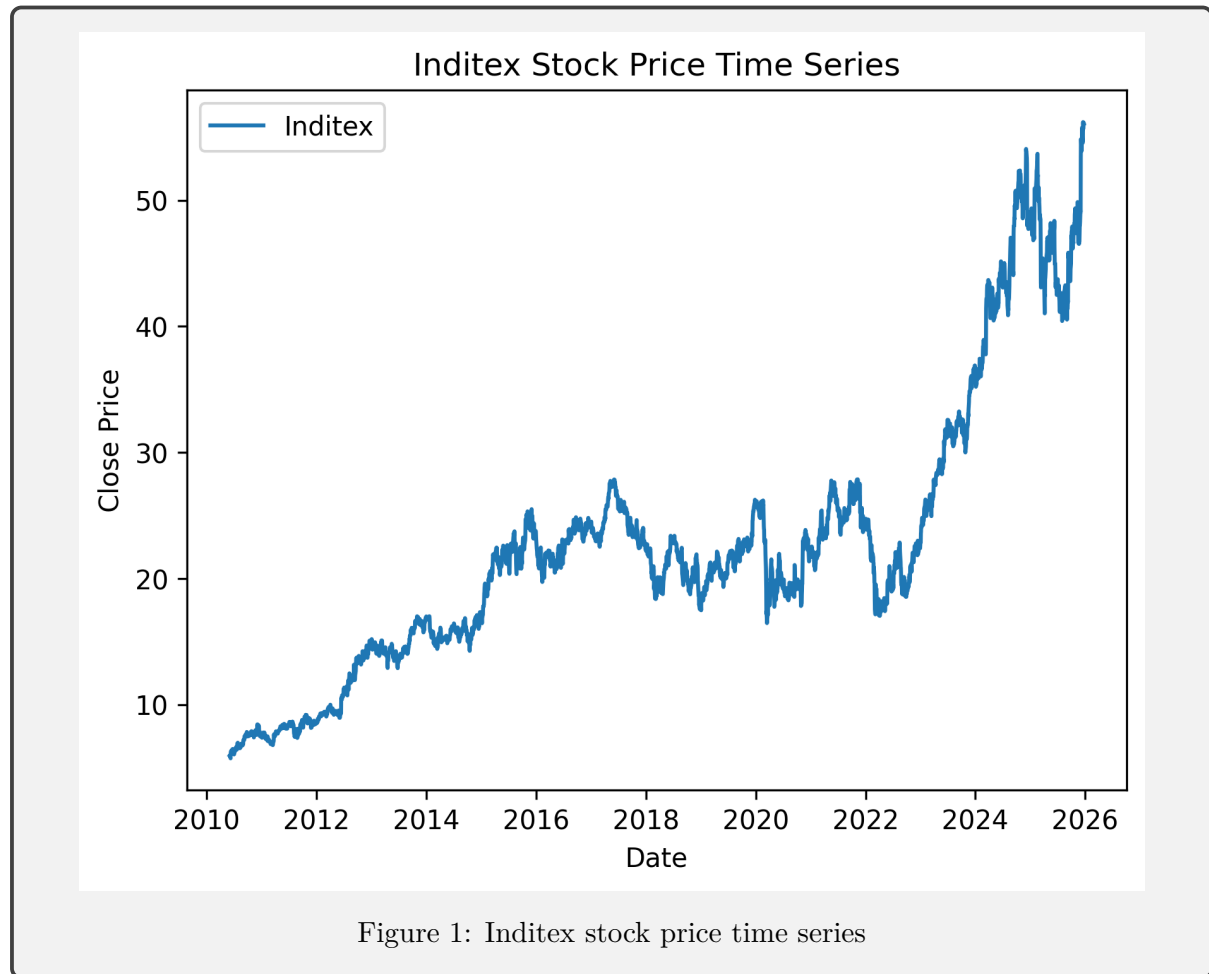
```
1 # Summary of Date intervals (75,296 total evaluated)
2 count          75296
3 mean      1 days 10:14:21.368465
4 std        0 days 20:18:55.691206
5 min          1 days 00:00:00
6 25%          1 days 00:00:00
7 50%          1 days 00:00:00
8 75%          1 days 00:00:00
9 max          5 days 00:00:00
10 Name: Date, dtype: object
```

3 Exploratory Data Analysis

The plots reveal long-term trends and volatility fluctuations typical of financial markets.

3.1 Stock Price Time Series

The historical closing prices reflect the long-term evolution and structural trends of the IBEX-35 components. Figure 1 highlights the price trajectory of Inditex.



The plots reveal sustained growth periods interspersed with volatility clusters typical of European equity markets.

3.2 Daily Returns Analysis

Financial analysis is conducted on daily log-returns to ensure stationarity. Returns were computed using the standard percentage change formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (1)$$

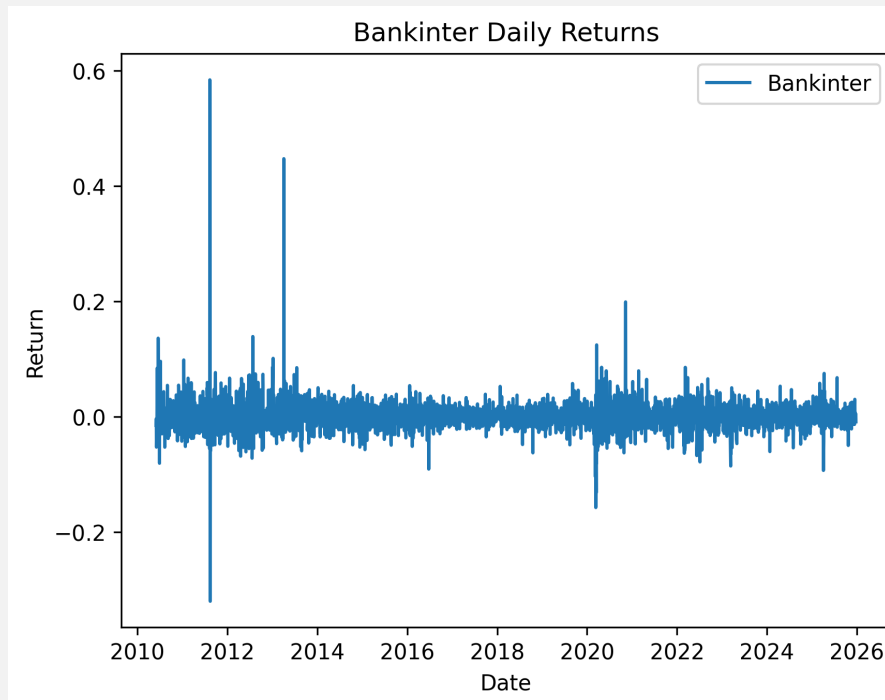


Figure 2: Bankinter daily returns: Visualizing volatility clustering

3.3 Statistical Profile of the IBEX-35

The descriptive statistics below provide a quantitative snapshot of the return distributions across 19 stocks.

Table 1: Detailed Descriptive Statistics of Daily Returns

Stock	Mean	Std	Skewness	Kurtosis	Min	Max
ACS	0.0007	0.0193	0.5102	17.6553	-0.1887	0.2585
Acciona	0.0006	0.0193	-0.2477	5.2612	-0.1749	0.1490
ArcelorMittal	0.0003	0.0264	0.0963	4.0053	-0.1818	0.1828
BBVA	0.0007	0.0212	-0.0087	5.6176	-0.1618	0.1665
B. Sabadell	0.0006	0.0253	0.3571	8.4976	-0.1929	0.2459
B. Santander	0.0005	0.0211	-0.1358	7.0105	-0.1989	0.1922
Bankinter	0.0011	0.0246	4.4905	116.1698	-0.3199	0.5840
Enagás	0.0004	0.0134	-0.5190	6.4204	-0.1480	0.0859
Ferrovial	0.0009	0.0151	-0.0966	6.2915	-0.1248	0.1362
Grifols	0.0005	0.0219	-1.1779	28.1619	-0.3493	0.1970
Iberdrola	0.0006	0.0139	-0.3241	7.0675	-0.1406	0.1008
Inditex	0.0007	0.0161	0.3746	5.0858	-0.1053	0.1403
Indra	0.0006	0.0210	0.4502	7.4369	-0.1476	0.1915
IAG	0.0006	0.0265	-0.3829	11.5750	-0.2790	0.2607
Mapfre	0.0006	0.0181	0.7261	8.8521	-0.1103	0.1997
Naturgy	0.0005	0.0153	-0.4982	12.1226	-0.1669	0.1548
Red Eléctrica	0.0005	0.0132	-0.3343	4.9155	-0.1338	0.0713
Repsol	0.0004	0.0194	0.3076	7.9845	-0.1513	0.1823
Telefónica	0.0000	0.0159	-0.0549	12.8533	-0.1612	0.1780

Key Statistical Observations

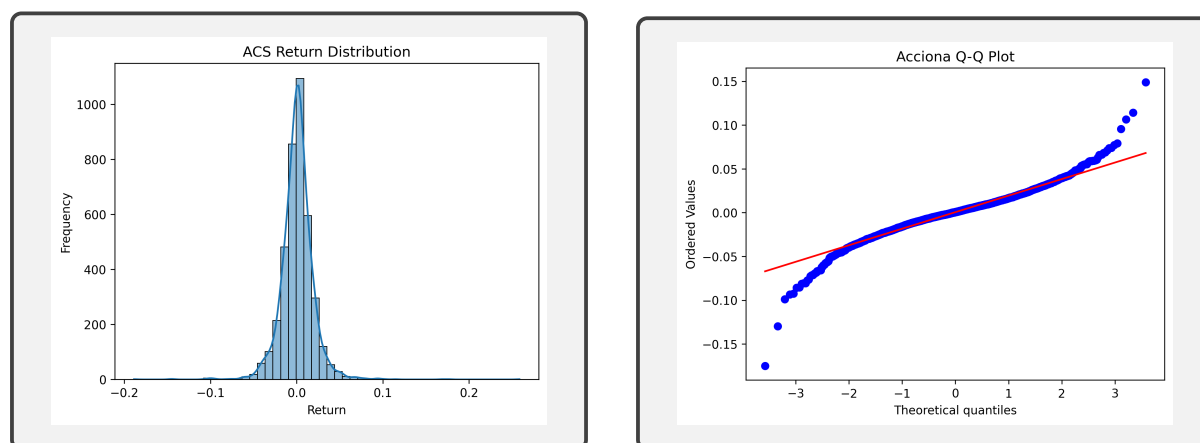
- **Outlier Alert:** *Bankinter* exhibits an extreme Kurtosis of 116.17, indicating a distribution with exceptionally heavy tails compared to the Gaussian norm.
- **Volatility Leader:** *IAG* and *ArcelorMittal* demonstrated the highest risk levels with standard deviations of $\sim 2.6\%$.
- **Conservative Assets:** *Red Eléctrica* and *Enagás* were identified as the most stable stocks, maintaining a standard deviation near 1.3% .

4 Normality Tests

A critical assumption in many financial models is the normality of returns. To evaluate whether the IBEX-35 daily returns follow a Gaussian distribution, we employed both visual diagnostics and formal statistical inference.

4.1 Visual Diagnostics: Histograms and Q-Q Plots

The empirical distribution of returns was compared against a theoretical normal curve. Figure 3 illustrates the distribution for the selected assets.



(a) Histogram: Returns vs. Normal Curve

(b) Q-Q Plot: Empirical vs. Theoretical

Figure 3: Visual Normality Assessment for IBEX-35 Stocks

The Histogram reveals a high peak (leptokurtosis) and "fat tails," indicating that extreme events occur more frequently than predicted by a normal distribution. The Q-Q Plot shows significant deviations from the 45-degree reference line at both extremities, confirming non-normal behavior.

4.2 Statistical Inference

To quantify the departure from normality, we performed the Jarque-Bera and Shapiro-Wilk tests on the combined universe of 19 stocks (75,000 observations).

Formal Normality Test Results

Jarque-Bera Test

(Tests Skewness and Kurtosis)

Statistic: 2,020,500.46

p-value: 0.0

Shapiro-Wilk Test

(Tests Correlation with Normal Distribution)

Statistic: 0.9058

p-value: 1.8177e-104

Conclusion on Normality

For all evaluated stocks, the p-values are functionally zero ($p < 0.05$). We **strongly reject the null hypothesis** (H_0) that the returns follow a normal distribution. The presence of high kurtosis (up to 116.16) confirms that the IBEX-35 exhibits heavy-tailed behavior, which is a standard characteristic of empirical financial time-series data.

5 Correlation Analysis

A pairwise correlation matrix was computed to identify the inter-relationships between stock returns, which is essential for understanding diversification potential within the IBEX-35.

5.1 Correlation Matrix

Raw Data Access

The full 19×19 correlation matrix contains detailed coefficients for all stock pairs but attaching or mentioning as table would be messy but you can see on below:

Click here to view CorrMatrix.csv

5.2 Sectoral Clusters and Diversification

The analysis reveals high internal correlation within the banking and utility sectors, while healthcare (Grifols) acts as a significant diversifier.

Table 2: Key Correlation Pairs (Rounded to 2 Decimals)

Stock 1	Stock 2	Correlation	Sector Insight
Banco Santander	BBVA	0.87	Banking Cluster (Highest)
Banco Sabadell	Banco Santander	0.72	Banking Cluster
Red Eléctrica	Enagás	0.68	Utility Cluster
Grifols	Bankinter	0.16	Diverse Pair (Lowest)
Repsol	Grifols	0.17	Diverse Pair
ArcelorMittal	Grifols	0.19	Diverse Pair

Diversification Summary

- **High Dependency:** The highest observed correlation is between *Banco Santander* and *BBVA* (0.87). This highlights a strong systemic risk within the Spanish financial sector.
- **Optimal Diversifiers:** *Grifols* consistently shows the lowest correlation magnitudes against all other sectors, particularly against *Bankinter* (0.16) and *Repsol* (0.17). Including such low-correlation assets is key to reducing overall portfolio variance.
- **Systemic Utility Link:** *Red Eléctrica* and *Enagás* maintain a robust positive correlation of 0.68, reflecting their shared sensitivity to energy regulations.

5.3 Correlation Heatmap

The heatmap in Figure 4 provides a global view of the systemic dependencies within the index.

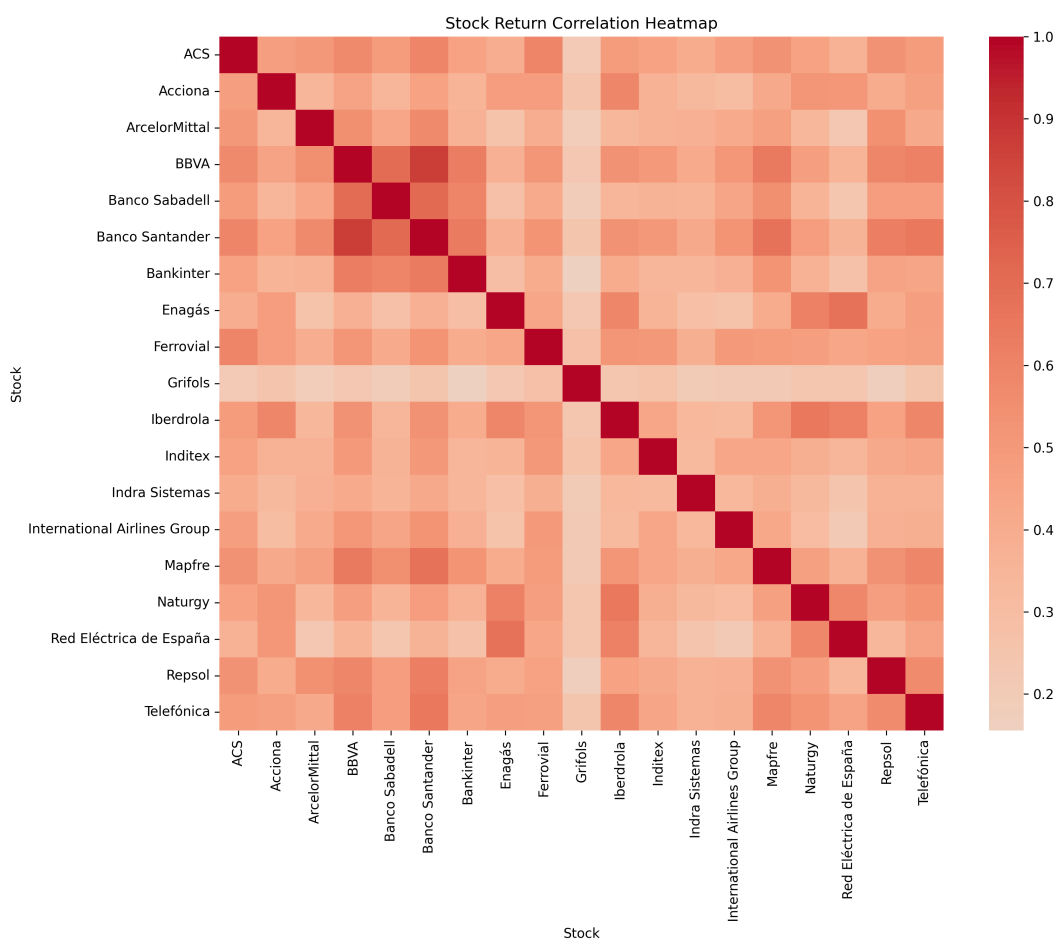


Figure 4: Heatmap of Pairwise Correlations: Identifying Systemic Risk Clusters

Most correlations are positive, indicating that stocks generally move in tandem with broader market trends. However, the intensity of these relationships varies significantly by sector.

6 Random Portfolio Generation

To visualize the relationship between risk and return, we constructed a two-asset portfolio universe utilizing **Inditex** and **Iberdrola**. This simulation demonstrates the power of asset allocation in determining a portfolio's risk-return profile.

6.1 Simulation Methodology

A total of 5,000 portfolios were simulated using randomly assigned weights w_i , such that $\sum_{i=1}^n w_i = 1$. For each iteration, the portfolio return (μ_p) and portfolio risk (σ_p) were calculated using the following matrix notation:

$$\mu_p = w^T \mu \quad (2)$$

$$\sigma_p = \sqrt{w^T \Sigma w} \quad (3)$$

Where μ represents the vector of expected returns and Σ is the covariance matrix of the assets.

```

1 for i in range(N):
2     w = np.random.random(2)
3     w = w / np.sum(w)
4
5     ret = np.dot(w, meanRet)
6     risk = np.sqrt(np.dot(w.T, np.dot(cov, w)))
7
8     sharpe = ret / risk
9     results.append([ret, risk, sharpe, w[0], w[1]])
10
11 portfolios = pd.DataFrame(results, columns=["Return", "Risk", "Sharpe",
12     "WgtInditex", "WgtIberdrola"])
13 portfolios.to_csv("Q5-Results/randPortfolios.csv", index=False)

```

6.2 Feasible Portfolio Region

The resulting scatter plot in Figure 5 maps the empirical feasible region.

Portfolio Analysis Observations

- **Efficient Bounds:** The scatter plot forms a parabolic "bullet" shape, where each point represents a unique allocation between the two stocks.
- **Risk Gradient:** The transition of points across the diagram illustrates how different weights influence the overall Sharpe Ratio and volatility of the investment.
- **Diversification Benefit:** The curvature of the left-most edge indicates that a specific blend of Inditex and Iberdrola results in a portfolio with lower risk than either asset held individually.

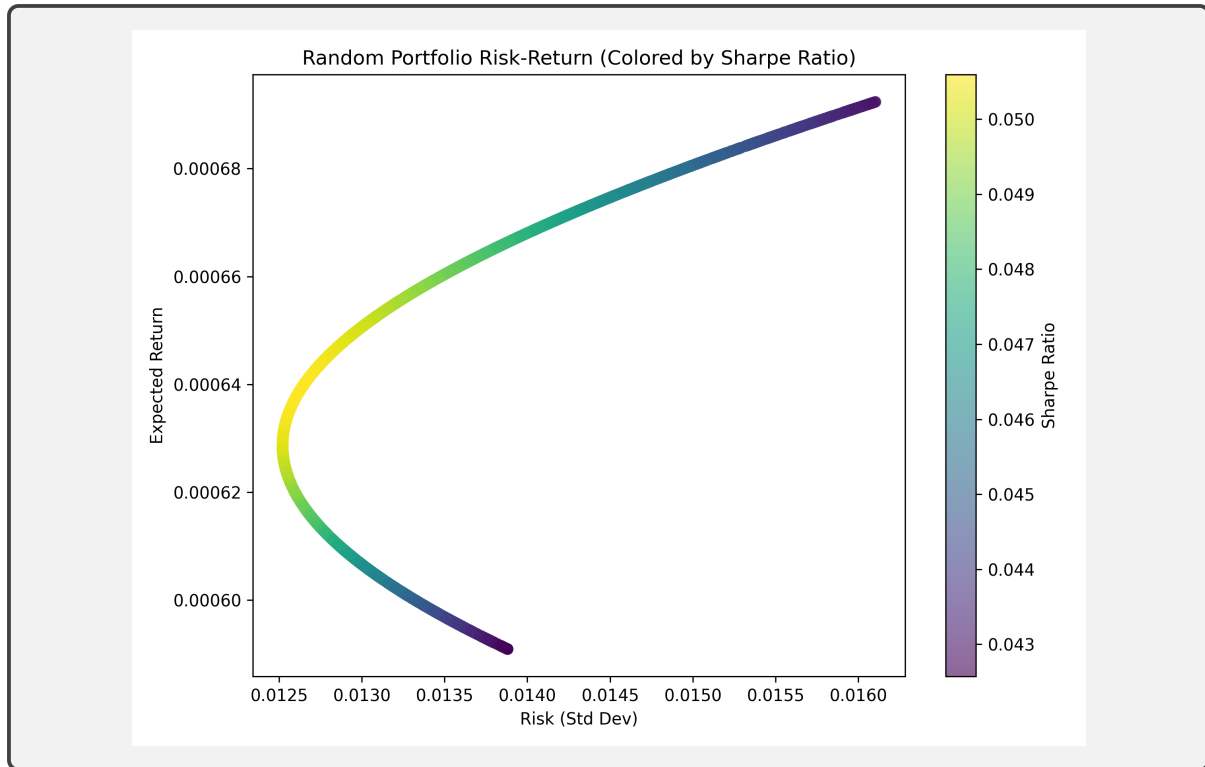


Figure 5: Risk-Return Scatter Diagram: Feasible Region for Inditex and Iberdrola

7 Markowitz Portfolio Optimization

Building upon the feasible region analysis, we applied Modern Portfolio Theory (MPT) to identify the set of optimal portfolios that offer the highest expected return for a defined level of risk.

7.1 Optimization Methodology

The optimization was conducted using the Sequential Least Squares Programming (SLSQP) constraint solver. We defined the objective function to minimize portfolio variance:

$$\min_w \sigma_p^2 = w^T \Sigma w \quad (4)$$

Subject to the following constraints:

- **Full Investment:** $\sum_{i=1}^n w_i = 1$
- **No Short Selling:** $w_i \geq 0$

7.2 Minimum Variance Portfolio (MVP)

The Minimum Variance Portfolio represents the specific point on the Efficient Frontier with the lowest historical standard deviation. Based on the optimization results for the **Inditex** and **Iberdrola** universe, the optimal weights are:

- **Inditex Allocation:** 37.02%
- **Iberdrola Allocation:** 62.98%

7.3 The Efficient Frontier

The Efficient Frontier was generated by solving for the minimum risk at varying levels of target returns.

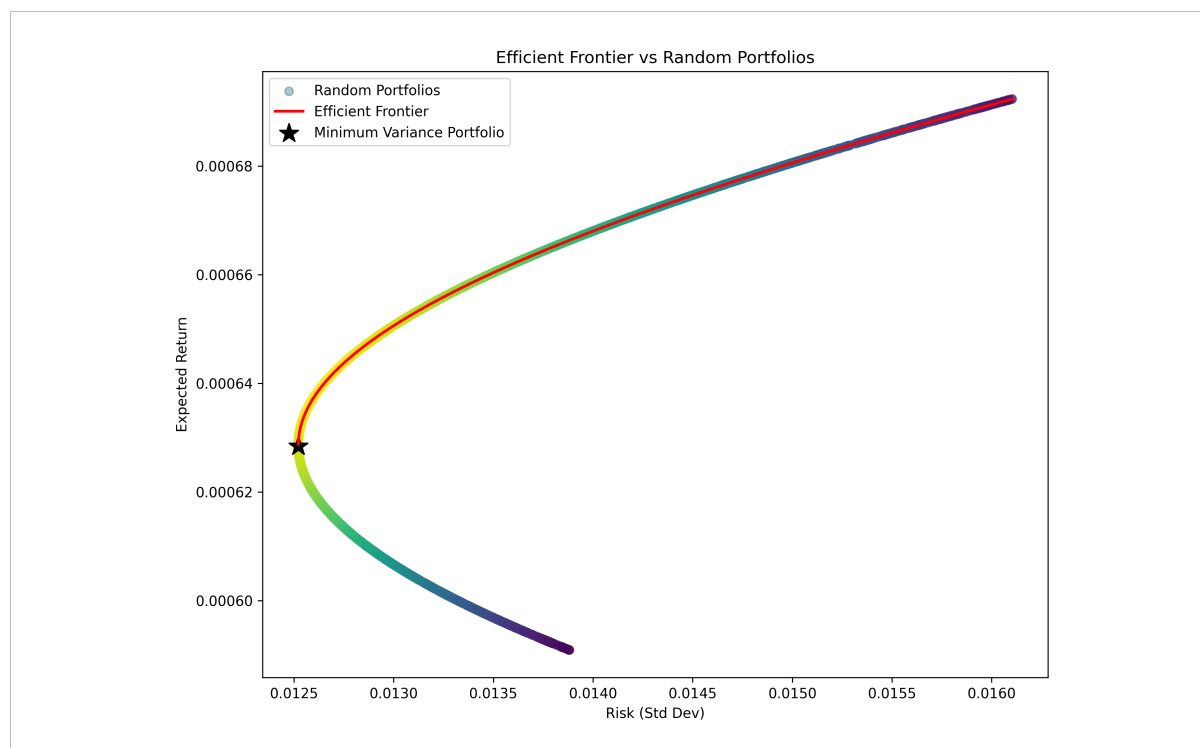


Figure 6: Markowitz Efficient Frontier: Random Portfolios and the Optimal Curve

As illustrated in Figure 6, the efficient frontier represents the upper boundary of the feasible region. The **Minimum Variance Portfolio** is clearly indicated at the "nose" of the curve, representing the most conservative allocation for an investor seeking to minimize volatility.

Final Observations

- **Dominance:** Any portfolio falling on the Efficient Frontier dominates the random portfolios located in the interior of the feasible region by providing higher returns for the same level of risk.
- **Asset Contribution:** Iberdrola's higher weight in the MVP (62.98%) is a direct result of its lower individual volatility and its covariance relationship with Inditex.

8 Observations

8.1 Distribution of Returns

The analysis shows that stock returns in the IBEX-35 do not follow a "Normal" (bell-shaped) curve. They show the HEAVY TAILS (around 4x-6x more than expected) which can be seen in the file **tailEvents.csv**.

8.2 Feasible Portfolio Region

When we plot thousands of random portfolios, they form a curved "bullet" shape. This happens because the two stocks, Inditex and Iberdrola, do not move perfectly together. By mixing them, we can actually cancel out some risk that you would have if you only held one stock.

8.3 Efficient Frontier

The top edge of the curved region is the **Efficient Frontier**. These are the "best" possible portfolios because they give you the highest return for every level of risk you take. Any portfolio below this line is considered "inefficient" because you could get a better return for the same risk elsewhere.

8.4 Minimum Variance Portfolio

The **Minimum Variance Portfolio (MVP)** is the point at the very tip of the curve. It represents the safest possible combination of assets. In our case, a mix of 37.02% Inditex and 62.98% Iberdrola gave us the lowest possible volatility.

9 Conclusion

This assignment shows how we can use Python and math to make better investment decisions. By looking at 15 years of IBEX-35 data, we proved that stocks are unpredictable and non-normal.

10 References

References

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